KERALA POLYTECHNIC • SEMESTER 4 • PROGRAM CORE

# Linear Integrated Circuits

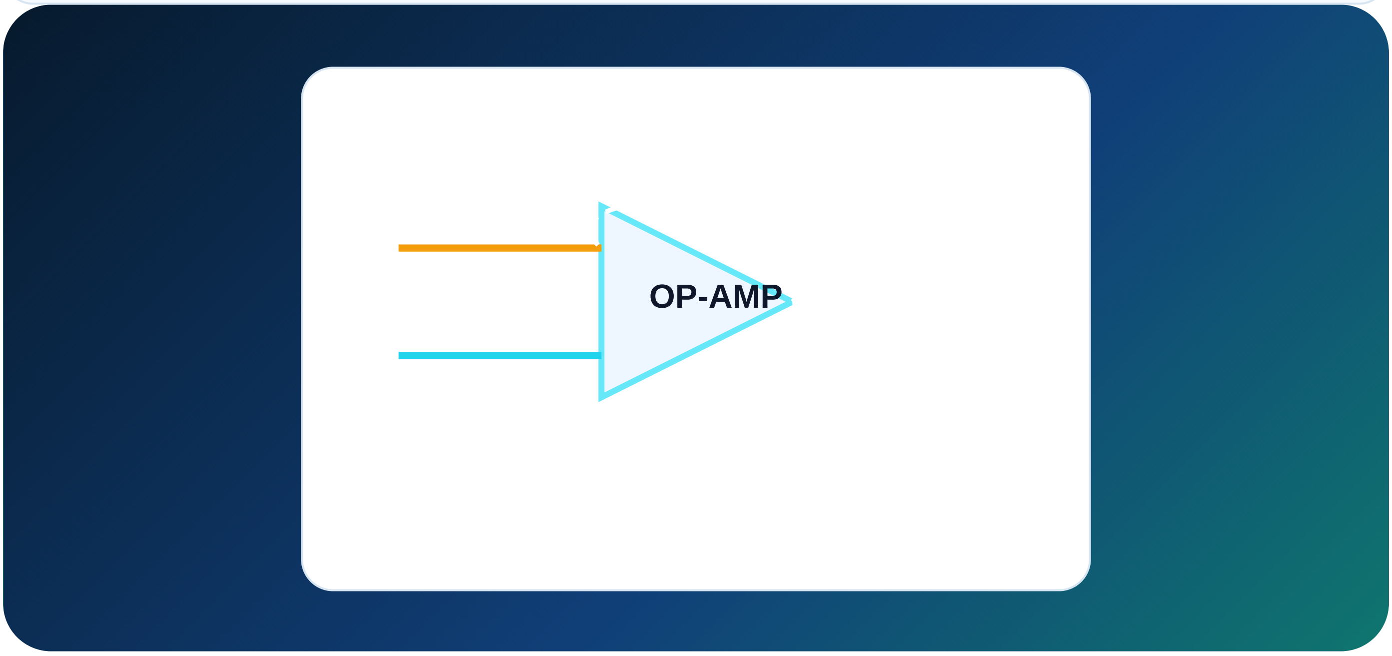
Complete handbook for Course 4043. It covers operational amplifier fundamentals, differential amplifier, feedback circuits, practical op-amp applications, 555 timer, PLL, IC voltage regulators, ADC and DAC. This page is made as a full-screen study handbook, not a short note.

COURSE CODE
**4043**

CREDITS
**4**

PERIODS
**60**

PATTERN
**L:3 T:1 P:0**



# Course identity and outcomes

## Official details

- Program: Diploma in Electronics, Electronics and Communication Engineering, Biomedical Engineering.
- Course title: Linear Integrated Circuits.
- Semester: 4.
- Course category: Program Core.
- Periods per week: 4.

## Professional use

Linear ICs are used in sensor signal conditioning, level detection, filtering, timing, regulated supplies and conversion between analog and digital systems. In escalator/electrical electronics, the same concepts appear in controller power supplies, sensor thresholds, speed or position feedback, temperature measurement, UPS/SMPS boards and embedded ADC inputs.

> Malayalam: Op-amp □□□ amplifier □□□□□□□ □□□□. Sensor signal correct level-□□□□□□□□ □□□□□□□□, noise □□□□□□□□□□□□, threshold detect □□□□□□□□, ADC-□□□□□ suitable signal □□□□□□ □□□□□□□□□□□□□□□□ main analog IC □□□.

## Course outcomes

| CO | Outcome | Hours | Level |
|---|---|---|---|
| CO1 | Illustrate internal structure, working and electrical parameters of operational amplifier. | 12 | Understanding |
| CO2 | Develop circuits using operational amplifier. | 20 | Applying |
| CO3 | Implement circuits using timer IC and PLL. | 12 | Applying |
| CO4 | Explain fixed and variable regulators, ADC and DAC. | 14 | Understanding |
| Test | Series test | 2 | - |

# Official syllabus structure expanded

| Module | Syllabus contents | Hours | Handbook expansion |
|---|---|---|---|
| M1 | Differential amplifier, op-amp block diagram, DC and AC parameters, equivalent circuit, transfer curve, feedback, inverting, non-inverting and follower circuits. | 12 | Fundamental theory, parameters, circuit behavior and exam notes. |
| M2 | Summing, difference, scaling, averaging, instrumentation, converters, integrator, differentiator, rectifier, detector, clipper, clamper, log, antilog, oscillators, multivibrators, wave generators, comparators and active filters. | 20 | Application circuits, design formulas and troubleshooting points. |
| M3 | 555 timer functional diagram, astable, monostable, PLL, IC 565, lock range, capture range, AM/FM detection and frequency control applications. | 12 | Timer formulas, PLL operation and block diagrams. |
| M4 | 78XX, 79XX, LM723, current boosting, current limiting, foldback protection, optocoupler, weighted/R-2R DAC, ADC types and sample-and-hold. | 14 | Power supply and data converter handbook. |

# Module 1 - Fundamentals of operational amplifier

## Differential amplifier

A differential amplifier amplifies the difference between two input signals and rejects the common signal. It is the input stage of an op-amp because practical sensor signals often contain common-mode noise.

$$V_{out} = A_d(V_1 - V_2) + A_c(V_{common})$$

$$CMRR = A_d / A_c \text{ and } CMRR \text{ in dB} = 20 \log_{10}(A_d/A_c)$$

- Balanced input, balanced output gives best differential operation.
- Single-ended output is common in IC output stages.
- High CMRR is required for low-noise measurement.

## Op-amp internal block

A practical op-amp has differential input stage, high-gain voltage amplifier, level shifting stage and output driver. The input stage gives high input resistance. The output stage gives low output resistance.

| Block | Function |
|---|---|
| Differential input | Accepts + and - inputs and rejects common signal. |

| | |
|---|---|
| Gain stage | Provides most of the voltage gain. |
| Level shifter | Sets DC level for proper output swing. |
| Output stage | Drives load with low output resistance. |

### Important parameters

- Input offset voltage: input voltage needed to make output zero.
- Input bias current: average current entering input pins.
- Slew rate: maximum rate of change of output voltage.
- CMRR: common-mode rejection capability.
- PSRR: power supply rejection capability.
- Gain bandwidth product: product of gain and bandwidth.

### Feedback and basic circuits

Negative feedback reduces gain but increases stability, bandwidth and accuracy. With high open-loop gain, closed-loop gain is mainly decided by external resistor ratio.

Inverting gain $A_v = -R_f/R_{in}$

Non-inverting gain $A_v = 1 + R_f/R_1$

Voltage follower gain $A_v = 1$

Voltage follower gain 1 □□□□□□□□□□□ important □□□. Sensor load □□□□□□□□□ next circuit-□□□□□□□ signal buffer □□□ □□□□□□□□□.

## Module 2 - Applications of operational amplifier

### Summing, difference and instrumentation amplifiers

Summing amplifier adds multiple input signals with selected weights. Difference amplifier gives output proportional to V2 - V1. Instrumentation amplifier gives high input impedance, high CMRR and accurate low-level signal amplification.

Summing: $V_o = -R_f(V_1/R_1 + V_2/R_2 + ...)$

Difference: $V_o = (R_2/R_1)(V_2 - V_1)$

Instrumentation amplifiers are used in strain gauge, load cell, biomedical sensor and industrial measurement circuits.

## Converters, integrator and differentiator

Voltage-to-current converter produces current proportional to input voltage. Current-to-voltage converter is used with current-output sensors and photodiodes.

V-I converter: $I_{out} \approx V_{in}/R$

I-V converter: $V_{out} = -I_{in} R_f$

Integrator: $V_o = -(1/RC) \int V_i \, dt$

Differentiator: $V_o = -RC \, dV_i/dt$

## Wave shaping and nonlinear circuits

- Precision rectifier rectifies signals below ordinary diode cut-in voltage.
- Peak detector stores maximum input value on a capacitor.
- Clipper limits waveform above or below selected level.
- Clamper shifts DC reference of waveform.
- Log and antilog amplifiers use diode/transistor exponential characteristics.

## Oscillators, generators and filters

- Phase shift oscillator and Wien bridge oscillator generate sine waves.
- Astable multivibrator generates continuous square wave.
- Monostable multivibrator generates one pulse after trigger.
- Triangular and sawtooth wave generators use comparator and integrator action.
- Active low-pass, high-pass and band-pass filters use op-amp and RC network.

Wien bridge oscillator frequency $f = 1/(2\pi RC)$

First-order cutoff frequency $f_c = 1/(2\pi RC)$

### Comparator and Schmitt trigger

A comparator switches output high or low by comparing input with a reference. Schmitt trigger uses positive feedback and gives two thresholds. This hysteresis prevents false switching when the input is noisy or slowly varying.

Hysteresis = Upper trigger point - Lower trigger point

Engineering use: sensor threshold detection, zero-crossing detection, protection circuits and wave shaping.

## Module 3 - Timer IC 555 and PLL

### 555 timer block and working

The 555 timer contains a voltage divider, two comparators, SR latch, discharge transistor and output driver. The internal references are 1/3 Vcc and 2/3 Vcc.

- Trigger pin below 1/3 Vcc sets output high.
- Threshold pin above 2/3 Vcc resets output low.
- Discharge transistor controls capacitor discharge.

### 555 monostable and astable

Monostable has one stable state and gives a single output pulse after trigger. Astable has no stable state and generates continuous square wave.

Monostable pulse width $T = 1.1RC$

Astable $T_{on} = 0.693(R_A + R_B)C$

Astable $T_{off} = 0.693 R_B C$

Frequency $f = 1.44 / ((R_A + 2R_B)C)$

### PLL operation

A phase locked loop locks the phase and frequency of a voltage-controlled oscillator to the input signal. Main blocks are phase detector, low-pass filter and VCO.

- Lock range: range where PLL remains locked.
- Capture range: range where PLL can acquire lock.
- Pull-in time: time required to reach locked condition.
- PLL IC 565 is used for laboratory study of PLL operation.

### PLL applications

PLL is used for AM and FM detection, frequency multiplication, frequency division and frequency synthesizing. In communication and measurement systems it helps track frequency changes and recover information from modulated signals.

## Module 4 - Voltage regulators, ADC and DAC

### Fixed regulators 78XX and 79XX

78XX series gives positive fixed output voltage and 79XX series gives negative fixed output voltage. Examples: 7805 gives +5 V, 7812 gives +12 V and 7905 gives -5 V.

Practical check: output voltage alone is not enough. Also check input voltage, load current, dropout margin, heat dissipation and capacitor placement.

> Regulator power loss $P = (V_{in} - V_{out}) I_{load}$

### LM723 and protection

LM723 is an adjustable voltage regulator IC. It has reference source, error amplifier, series pass control and protection options.

- Current boosting uses external pass transistor.
- Current limiting protects against overload.
- Foldback protection reduces short-circuit current.

- Optocoupler gives isolation using light.

### Digital to analog converters

A DAC converts binary input into proportional analog output. Specifications include resolution, accuracy, settling time, linearity and monotonicity.

- Weighted resistor DAC uses binary weighted resistor values.
- R-2R ladder DAC uses only R and 2R values and is easier for IC implementation.

$$\text{Number of levels} = 2^n$$

### Analog to digital converters

An ADC converts analog input to digital code. Sample-and-hold holds the input constant during conversion.

| ADC | Speed | Key point |
|---|---|---|
| Flash | Very high | Uses many comparators |
| Counter ramp | Low | Simple counter and DAC feedback |
| SAR | Medium/high | Binary search method |
| Dual slope | Low | High noise rejection, used in DMM |

$$\text{Approx. ADC step size} = V_{ref} / 2^n$$

## Formula bank

### Op-amp

$$A_{v\,inv} = -R_f/R_{in}$$

$$A_{v\,non\text{-}inv} = 1 + R_f/R_1$$

CMRR dB = 20log10(Ad/Ac)

### 555 and filters

$T_{mono} = 1.1RC$

$f_{astable} = 1.44/((RA+2RB)C)$

$f_c = 1/(2\pi RC)$

### Converters

ADC levels = $2^n$

Step size ≈ $V_{ref}/2^n$

PD regulator = $(V_{in} - V_{out})I$

## Solved examples

### Example 1 - Inverting amplifier

Rin = 10 kΩ, Rf = 100 kΩ, Vin = 0.2 V. Gain = -100k/10k = -10. Output = -10 × 0.2 = -2 V.

### Example 2 - CMRR

Ad = 2000 and Ac = 0.2. CMRR = 10000. CMRR dB = 20 log10 10000 = 80 dB.

### Example 3 - 555 pulse

R = 100 kΩ and C = 10 µF. T = 1.1RC = 1.1 × 100000 × 10 × 10^-6 = 1.1 s.

### Example 4 - ADC step

For 8-bit ADC with Vref = 5 V, step size ≈ 5/256 = 0.0195 V = 19.5 mV.

## Expected questions and practice bank

### Module 1

1. Explain BJT differential amplifier.
2. Define and calculate CMRR.
3. Draw and explain op-amp block diagram.
4. Explain input offset voltage, bias current and slew rate.
5. Derive inverting amplifier gain.
6. Explain voltage follower.

### Module 2

1. Explain summing and difference amplifier.
2. Explain instrumentation amplifier.
3. Draw integrator and differentiator.
4. Explain precision rectifier and peak detector.
5. Explain Schmitt trigger.
6. Explain active filters.

### Module 3

1. Draw 555 block diagram.
2. Explain monostable operation.
3. Explain astable operation and formula.
4. Draw PLL block diagram.
5. Define lock range and capture range.
6. State PLL applications.

### Module 4

1. Explain 78XX and 79XX regulators.
2. Explain LM723 regulator.
3. Explain current limiting and foldback protection.
4. Compare weighted and R-2R DAC.
5. Compare flash, counter, SAR and dual-slope ADC.
6. Explain sample-and-hold.

## Model question paper

Course: 4043 Linear Integrated Circuits
Time: 3 Hours      Maximum Marks: 100

Part A - Short answers
1. Define CMRR.
2. State two ideal op-amp characteristics.
3. Write inverting amplifier gain equation.
4. What is virtual ground?

5. State use of voltage follower.
6. Define slew rate.
7. List two applications of 555 timer.
8. Define capture range of PLL.
9. Name two fixed positive regulators.
10. What is DAC resolution?

Part B - Medium answers
11. Explain differential amplifier.
12. Explain op-amp internal block diagram.
13. Explain summing and difference amplifiers.
14. Explain integrator and differentiator.
15. Explain Schmitt trigger and hysteresis.
16. Explain 555 monostable circuit.
17. Explain PLL operation.
18. Compare flash ADC and SAR ADC.

Part C - Long answers
19. Explain inverting and non-inverting amplifiers with gain derivation.
20. Explain precision rectifier, peak detector, clipper and clamper.
21. Explain phase shift and Wien bridge oscillators.
22. Explain 555 astable multivibrator with timing equations.
23. Explain LM723 regulator, current limiting and foldback protection.
24. Explain DAC types and ADC classification.

# Answer key

## Part A key

1. Ratio of differential gain to common-mode gain.
2. Infinite input resistance, zero output resistance, infinite gain, infinite bandwidth.
3. Av = -Rf/Rin.
4. Inverting node stays nearly at ground potential due to negative feedback.
5. Buffering and impedance matching.
6. Maximum rate of output voltage change.
7. Delay, pulse generation, oscillator, PWM.
8. Input frequency range where PLL can acquire lock.
9. 7805, 7812.
10. Smallest output change for one digital count.

## Long answer direction

For every circuit answer, draw neat diagram, mark input and feedback path, write formula, explain operation and mention application. For comparison answers, use a table. For regulator questions, include input, reference, error amplifier, pass transistor and protection.

> Exam-□ diagram + formula + working + application □□□□ order follow □□□□□□□ answer standard □□□ □□□□□□.

## References

Roy D. C. and S. B. Jain, Sergio Franco, R. A. Gayakwad, Botkar, Salivahanan and David A. Bell are the main references listed for this course. Use Gayakwad for easier diploma-level understanding and Sergio Franco for deeper design logic.

**No-nonsense study advice:** Do not memorize only circuit names. Learn the feedback path and formula. In Linear IC, marks come from circuit diagram, operating principle and correct relation between resistor/capacitor values and output.